Assignment - 10

Vehicle Service Request Form (Advanced) — Advanced Fields Use Case

1. Objective

To extend the existing "Vehicle Service Request Form" application in Zoho Creator by incorporating all seventeen Advanced Fields available in the Form Builder, so that the form moves beyond basic data capture and demonstrates automation, computed values, file/media handling, and structured lookups within a real automobile service-center use case.

2. Problem Statement

The basic Service Request Form (built earlier using the thirteen Basic Fields) captures a vehicle owner's identity, contact details, and issue description, but it cannot handle several real-world requirements of a growing service center: verifying the vehicle model and standard service package against a master vehicle-model list, reviewing the vehicle's previous service history in one submission, computing the net service cost after discounts, issuing a unique scannable service tracking reference, or accepting supporting media such as an insurance copy, a video of the reported issue, or a signed service authorization.

Extend the Service_Request_Form in Zoho Creator so that, in addition to the existing basic fields, it uses every Advanced Field type provided by Zoho Creator's Form Builder to support service history review, cost computation, vehicle model verification, and automated service-ID/QR generation for tracking through the repair pipeline.

3. Scope of the Use Case

- Application Name: Vehicle Service & Repair Request Management (Advanced Edition)
- Form Name: Service_Request_Form (extended with Advanced Fields)
- Primary Users: Vehicle owners / Service Center Front Office and Technicians
- Report View: Service_Request_Report (extended with computed cost, service request ID, and QR code columns)
- Objective of the Extension: Enable the service center to verify vehicle models automatically, review service history, compute net payable service costs, and issue scannable digital service tracking references

4. Advanced Fields Used and Their Mapping

The table below maps every Advanced Field from the Zoho Creator Form Builder palette to its role in the extended Vehicle Service Request Form.

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Decision Box	Requires_Pickup_Drop_Service	Captures a quick Yes/No decision on whether the owner needs the vehicle picked up from and dropped back to their address	Yes / No toggle

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Rich Text	Service_Center_Guidelines_Display	Displays formatted service center guidelines (bold headings, bullet points, hyperlinks) directly within the form	"Please remove personal belongings before drop-off. Estimates are shared before any repair work begins."
URL	Insurance_Policy_Portal_URL	Captures a link to the owner's online motor insurance policy page, for cashless service claims	https://insureportal.example.com/policy/91223
Image	Vehicle_Damage_Photo	Captures a photograph of the reported damage or fault area on the vehicle for preliminary assessment	damage_photo_02.jpg
Percent	Service_Discount_Percent	Captures the discount percentage applicable for loyalty customers or ongoing service offers	10%
Currency	Estimated_Service_Cost	Captures the estimated cost of the service or repair, formatted with currency symbol	₹4,200.00
Decimal	Vehicle_Age_Years	Captures the age of the vehicle in years, used to determine applicable warranty or wear-and-tear norms	3.5
Date-Time	Service_Request_Submitted_On	Automatically records the exact date and time the service request was submitted	24-Jul-2026 07:30
SubForm	Previous_Service_History	Captures repeating rows of previous service date, type of service, and cost incurred	Row 1: 12-Jan-2026 – Oil Change – ₹1,800; Row 2: 05-Apr-2026 – Brake Pad Replacement – ₹2,500
Lookup	Vehicle_Model_Lookup	Looks up and auto-fills the vehicle model's standard service package and estimated duration from a master Vehicle Models table	Hyundai i20 – Standard Service Package – 3 hours

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Signature	Owner_Digital_Signature	Captures the owner's digital signature authorizing the service center to proceed with the estimated work	Signed (drawn signature captured)
File Upload	Insurance_Copy_Upload	Allows the owner to upload a scanned copy of the vehicle's insurance document for cashless claim processing	insurance_copy.pdf
Audio	Engine_Sound_Voice_Note	Allows the owner to optionally record a short audio clip of an unusual engine or brake sound for the technician's reference	engine_sound.mp3 (0:18)
Video	Vehicle_Issue_Video	Allows the owner to upload a short video showing the reported issue, such as a warning light or unusual vibration	issue_video.mp4 (0:40)
Auto Number	Service_Request_ID	Automatically generates a unique, sequential service request number for every submission	SVC-REQ-2026-000691
Formula	Net_Service_Cost_Payable	Automatically computes the net service cost payable after applying the applicable discount	= Estimated_Service_Cost – (Estimated_Service_Cost × Service_Discount_Percent)
QR/Barcode	Service_Tracking_QR_Code	Automatically generates a scannable QR code encoding the Service Request ID for tracking through the repair pipeline	QR image encoding SVC-REQ-2026-000691

5. Extended Form Layout (Section-wise)

The sections below are added to the existing Owner, Contact, and Vehicle/Issue Details sections of the basic Vehicle Service Request Form.

Section E — Vehicle Verification & Evidence

- Vehicle Model Lookup (Lookup field, auto-fills from master Vehicle Models table)
- Vehicle Damage Photo (Image field)
- Insurance Copy Upload (File Upload field)
- Vehicle Age in Years (Decimal field)

Section F — Service History & Media Details

- Previous Service History (SubForm field, repeating rows)
- Insurance Policy Portal URL (URL field)
- Vehicle Issue Video (Video field, optional)
- Engine Sound Voice Note (Audio field, optional)

Section G — Cost & Pickup Preference

- Requires Pickup/Drop Service (Decision Box)
- Estimated Service Cost (Currency field)
- Service Discount Percent (Percent field)
- Net Service Cost Payable (Formula field, read-only, auto-computed)

Section H — Authorization & System-Generated Values

- Service Center Guidelines Display (Rich Text field, read-only reference block)
- Owner Digital Signature (Signature field)
- Service Request Submitted On (Date-Time field, auto-captured)
- Service Request ID (Auto Number field, system-generated)
- Service Tracking QR Code (QR/Barcode field, system-generated from Service Request ID)

6. Validation & Configuration Rules

- Lookup field (Vehicle Model Lookup): connected to a master Vehicle Models form; front office staff type-ahead search and select rather than typing free text.
- SubForm (Previous Service History): configured to allow a minimum of 0 and a maximum of 10 rows per vehicle.
- File Upload (Insurance Copy Upload): restrict accepted formats to PDF, JPG, and PNG, with a maximum size of 5 MB.
- Image (Vehicle Damage Photo): restrict accepted formats to JPG and PNG only, with a maximum size of 2 MB.
- Video (Vehicle Issue Video): restrict maximum duration to 45 seconds and maximum size of 50 MB.
- Percent (Service Discount Percent): restrict value range from 0 to 25.
- Currency (Estimated Service Cost): restrict to non-negative values only, formatted in INR (₹).
- Formula (Net Service Cost Payable): set as a read-only field, recalculated automatically whenever Estimated Service Cost or Service Discount Percent changes.
- Auto Number (Service Request ID): configured with prefix "SVC-REQ-", the current year, and a 6-digit auto-incrementing sequence.

- QR/Barcode (Service Tracking QR Code): configured to auto-generate immediately after the Service Request ID is assigned on submission.
- Signature (Owner Digital Signature) and Decision Box (Requires Pickup/Drop Service) are mandatory fields; the form cannot be submitted without them.

7. Workflow Summary

- Step 1: Owner completes the existing basic-field sections (personal, contact, and vehicle/issue details).
- Step 2: Front office verifies the Vehicle Model using the Lookup field, which auto-fills the standard service package and estimated duration.
- Step 3: The vehicle's Previous Service History is reviewed or added as rows in the SubForm for technician reference.
- Step 4: Owner uploads the insurance copy, damage photo, and optionally an issue video and a voice note of any unusual sound.
- Step 5: Owner indicates whether pickup/drop service is required, and the Estimated Service Cost, Service Discount, and computed Net Service Cost Payable are displayed.
- Step 6: Owner reviews the Service Center Guidelines rich text block and provides their Digital Signature to authorize the estimated work.
- Step 7: On Submit, Zoho Creator auto-generates the Service Request ID, timestamps the Service Request Submitted On field, and generates the Service Tracking QR Code.
- Step 8: The service center reviews the Service_Request_Report, verifies uploaded documents and service history, confirms the payable cost against Net Service Cost Payable, and uses the QR code to track the vehicle through each repair stage.

8. Expected Learning Outcomes

- Understand the purpose and correct use case for each Advanced Field in Zoho Creator.
- Learn how to configure Lookup fields for cross-form data verification.
- Practice using SubForms to capture repeating, service-history data within a single submission.
- Understand how Formula and Auto Number fields automate calculations and unique identifiers.
- Learn how to configure file-based fields (Image, File Upload, Audio, Video, Signature) with format and size restrictions.
- Understand how QR/Barcode fields can be used for real-world repair-tracking validation and process automation.

9. Deliverable for This Lab Exercise

Students must extend their existing Service_Request_Form in Zoho Creator to include all 17 Advanced Fields exactly as specified above, configure the listed validations and computed logic, and submit a screenshot-based lab record showing: (a) the extended form in Form Builder with all Advanced Fields, (b) a live preview of the form including the SubForm and computed Net Service Cost Payable, (c) at least two sample submitted records showing the auto-generated Service Request ID and QR Code in the report view, and (d) a screenshot of the Vehicle Model Lookup in action.